Fractionally Differentiated Features

Core dilemma: Stationarity vs Memory

ML algos need consistency, but also context to make predictions.

- Prices (Order of diff.: $d=0$): They have memory, but are non-stationary
- Returns (Order of diff.: $d=1$): They are memoryless, but are stationary

Solution: Fractional Differentiation

We can find the "sweet spot" for order of differentiation with $0<d<1$. Getting stationarity while keeping maximum of memory.

Mathematical Method

Introducing the Backshift operator B: $BX_t = X_{t-1}$, $B^2X_t = X_{t-2}$

- With $d=1$: $\Delta X_t = X_t - X_{t-1} \iff (1-B)X_t = X_t - X_{t-1}$
- With $d=2$: $(1-B)^2 = X_t - 2X_{t-1} + X_{t-2}$

If d is a fraction (ex: 0.4) you need to use the Binomial Theorem.
Therefore the expansion becomes an infinite series:

- With d fractional: $(1-B)^d = 1 - dB + \frac{d(d-1)}{2!}B^2 - \frac{d(d-1)(d-2)}{3!}B^3 + \dots$

- With $d=0.4$: ω_0 = always 1
  $\omega_1 = -0.4$ (current d)
  $\omega_2 = \frac{0.4(0.4-1)}{2!} = -0.12$
  $\omega_3 = -0.064$

⚠ If $d=1$ the formula stops after the second term because $(d-1)=0$.
If d is a fraction the weights decay asymptotically to zero (close but never reach it)

Dot Product

We can rewrite the fractionally diff. feature as a dot product of weights (ω) and historic values (X):

$$\tilde{X}_t = \sum_{k=0}^{\infty} \omega_k X_{t-k} \quad \text{with} \quad \omega_k = (-1)^k \prod_{i=0}^{k-1} \frac{d-i}{k!}$$

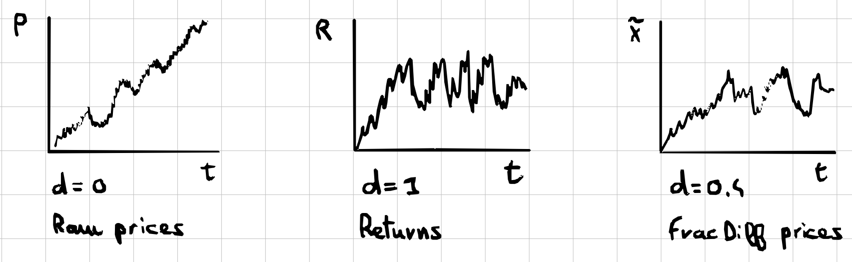
Implementation: Fixed-Width Fracdiff (FFD)

The expanding window method introduces a massive downward drift, solved by FFD.

Truncating negligible weights

Instead of letting weight window grow indefinitely, we set a threshold $\tau > 0$ for the weight modulus $|\omega_k|$. After that we discard all the smaller weights.

- Find index l^* where $|\omega_{l^*}| \geq \tau$ and $|\omega_{l^*+1}| < \tau$
- Vector of weights becomes: $\tilde{\omega}_k = \begin{cases} \omega_k & \text{if } k \leq l^* \\ 0 & \text{if } k > l^* \end{cases}$

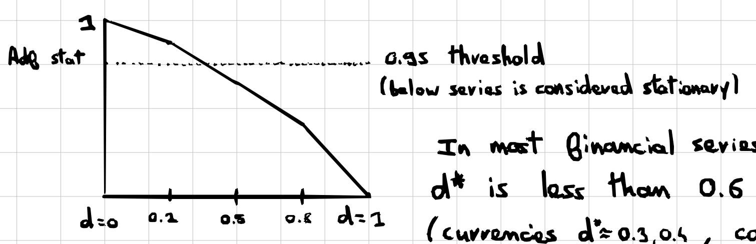


Optimal Frontier d^*

Let's define d^* as the min. coefficient of diff. required to make a non-stationary series become stationary.

- $d^* = 0$: original series is already stationary.
- $d^* < 1$: series contains unit root and requires some diff., most of the time $d^* \ll 1$.
- $d^* > 1$: explosive, compounding growth (ex: hyper-inflationary bubble).

ADF Test for stationarity



In most financial series d^* is less than $0.6 \ll 1$.
(currencies $d^* \approx 0.3, 0.4$, cattle $d^* = 0$)